

MIMO versus Decentralized Control of the Coupled Droop-Share and Drive Loops on the SC001 Balancer Board: An Internal Comparison

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Abstract: The SC001 scale car carries two independently designed robust loops: a shipped H_∞ /Youla-H current-share controller that commands a droop ratio between a fuel cell and a battery, and a vehicle-speed drive controller. Both were designed single-input single-output, on the assumption that the two loops do not interact. This report tests that assumption. We build a 2×2 small-signal design plant in which the drive channel disturbs the share channel through the bus current it draws, validate it against an independent 15-state converter-level model, synthesize a centralized 2×2 H_∞ controller with a matrix generalization of the Youla-H DC-correction recipe that makes $T(0) = I$ structural, and compare it head-to-head against the decentralized pair on the same coupled plant, the same corner families, and the same simulations. The coupling is one-directional, so the relative gain array is provably uninformative; the informative measure is the directional ratio $|G_{s,12}|/|G_{s,11}|$, which is 0.39 at the nominal 2 A load and 6.28 at 0.5 A. The comparison gives three different answers on the three headline metrics: the centralized controller wins on worst-corner sensitivity by roughly 20–25 %, ties on regen bus excursion because both controllers sit on the same current rail, and wins or loses on drive-transient share excursion depending on the sign of an uncalibrated source-voltage mismatch ΔV_0 . Decentralized control is stable at all 4992 feasible corners and is substantially better on every saturated large-signal case. We conclude that decentralization is justified rather than merely convenient, and that measuring ΔV_0 on the bench is the single highest-value action for narrowing the remaining uncertainty.

Keywords: H_∞ control; Youla parameterization; MIMO control; droop control; power sharing; decentralized control; hybrid vehicle; DC bus

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Internal lab report — not for external distribution or publication. Numbers are generated artefacts of the controller_design_MIMO pipeline; nothing here is bench-calibrated.

1. Introduction

The SC001 scale car’s DC balancer board routes power from a fuel cell and a battery onto a shared bus through two off-the-shelf boost regulators whose output resistance is made electronically programmable by injecting a shunt-derived, multiplying-DAC-scaled signal into each converter’s feedback node. The resulting *droop ratio* r is the actuator of a current-share loop whose output is the share fraction $\alpha = I_{FC}/(I_{FC} + I_{BT})$, closed by a robust H_∞ /Youla-H controller synthesized against a 60-corner uncertain plant family; it ships in firmware today and is documented separately [8]. The vehicle’s speed is regulated by a second loop that commands motor current through a VESC over UART, following the drive-controller recipe of [4] with the post-synthesis DC correction of [5].

Both loops were designed single-input single-output, each on its own plant, each validated only against its own uncertainty family. That is a decision, not a derivation — the two loops share a bus, and every ampere the motor draws is an ampere the two sources must split. The outlook section of [8] poses the question directly — *is the decentralized design actually justified, or merely convenient?* — and names three metrics on which a centralized design should be compared: worst-corner $\|S\|_\infty$, drive-transient share excursion, and regen-event bus excursion. This report answers that question.

A study of this shape can fail in two symmetric ways: it can be set up so the centralized controller cannot lose (evaluate the decentralized pair on a diagonal plant), or so it cannot win (compare on metrics that saturate the actuator). We avoid the first by closing *both* controllers against the same coupled 2×2 plant with the same assembly routine, corner sets and simulator; we address the second by stating the amplitude of every excitation and flagging every case in which the ± 5 A clamp is active.

Contributions.

- A 2×2 small-signal design plant for the coupled share/drive problem (Section 2), in which the drive-to-share coupling is *derived* from the static share law rather than asserted, and is reproduced endogenously to 0.032 % by an independent 15-state converter-level model.
- The observation that the plant is structurally upper-triangular, so the relative gain array is the identity at every frequency and every corner and carries no information — a methodological correction to the metric the outlook section proposed.
- A drive-channel H_∞ /Youla-H controller built by the recipe of [4,5], including a structural finding about that recipe's second-order sensitivity weight on a near-integrator mechanical plant (Section 3).
- A matrix generalization of the scalar Youla gain-tuning recipe that enforces $T(0) = I$ on a MIMO loop by a constant right multiplier on the Youla parameter, made *structural* by an exact rank-2 integrator split (Section 4). This is the report's control-theoretic contribution.
- A head-to-head comparison on identical terms (Section 6), whose result is that the three headline metrics give three different answers, and an independent MATLAB cross-validation of the whole chain (Section 7).

2. The Coupled 2×2 Plant

2.1. Notation

The share literature, the powertrain-Youla papers and the firmware reuse the same letters for different quantities. Table 1 is binding here; the three collisions that actually bite are r (droop ratio, never tire radius — that is r_t), α (current share, never an acceleration) and K (share-plant DC gain, never a controller gain or motor K_V).

Table 1. Notation. Small-signal deviations carry a leading Δ .

Symbol	Units	Meaning (and what it is <i>not</i>)
r	—	Droop ratio command, $r \in [0.15, 0.85]$. <i>Not</i> tire radius.
r_t	m	Tire radius, 0.033.
α	—	Current share $I_{FC}/(I_{FC} + I_{BT})$; the share-loop output. <i>Never</i> an acceleration.
k_d	Ω	Droop scale, 0.30 (firmware <code>K_DR00P</code>).
$R_{e,max}$	Ω	Full-scale programmable output resistance, 2.014 (as-built).
ΔV_0	V	No-load source-voltage mismatch, FC vs. BT; ± 0.4 budget, sign unknown.
I_{tot}	A	Total bus current out of the two sources.
K	—	Share-plant DC gain $\partial\alpha/\partial r$ at the operating point.
K_v	—	Drive-channel structural gain uncertainty multiplier, $\in \{0.5, 1, 2\}$.
k_t, φ, b_{eff}	—	Motor torque constant, drive ratio 9.49:1, wheel-referred damping.
G, G_s	—	Design plant; $G_s = D_e^{-1} G D_u$ its scaled form.
$S, T, Y; W_p, W_d, W_u$	—	Sensitivity, complementary sensitivity, Youla transfer function (2×2), and the H_∞ cost weights on each.
M, γ	—	MIMO Youla-H DC correction matrix $[G_s(0)Y_H(0)]^{-1}$; H_∞ level.

The plant inputs and outputs are

$$u = \begin{bmatrix} \Delta r \\ \Delta i_{cmd} \end{bmatrix}, \quad y = \begin{bmatrix} \Delta \alpha \\ \Delta v \end{bmatrix}, \quad (1)$$

with actuator limits $r \in [0.15, 0.85]$ and $|i_{cmd}| \leq 5$ A. Native rates are 1 kHz for the share loop and 500 Hz for the motor loop, the latter set by the VESC UART frame floor of roughly 781 μ s.

2.2. Share channel (recap)

The droop hardware realizes a commanded output resistance $R_e(g) = R_{e,max} g$ per channel; mapping the ratio as $R_F = k_d/r$, $R_B = k_d/(1-r)$ gives $R_F \parallel R_B = k_d$ independently of r , so the bus droop is decoupled from the split. Solving the two-Thévenin circuit yields the exact static share law [8]

$$\alpha(r, I_{tot}) = r + \frac{\Delta V_0 r(1-r)}{k_d I_{tot}}, \quad (2)$$

whose r -derivative is the share-plant DC gain

$$K = \frac{\partial \alpha}{\partial r} = 1 + \frac{\Delta V_0(1-2r_0)}{k_d I_{tot0}}. \quad (3)$$

The share channel of the design plant is then the shipped SISO plant unchanged,

$$G_{11}(s) = K \cdot \text{Padé}_2(e^{-T_d s}) \cdot \frac{1}{\tau_r s + 1} \cdot \frac{1}{\tau_f s + 1}, \quad (4)$$

with $T_d = 1$ ms nominal loop delay, $\tau_r = 100$ μ s, and $\tau_f = 0.8$ ms the firmware's 200 Hz share-measurement prefilter. A gate confirms G_{11} is identical to the shipped SISO plant to 8.0×10^{-16} relative over all operating points and five parameter corners.

Writing K as (3) rather than sweeping it as a free scalar makes the shipped SISO envelope $K \in [0.55, 1.45]$ *derived*. Inside the shipped domain ($I_{tot} \geq 2 \text{ A}$, $r \in [0.3, 0.7]$) the implied envelope is $[0.733, 1.267]$, comfortably inside; over the full feasible operating set it is $[0.533, 2.067]$, reaching 2.067 at light load with full mismatch (0.5 A , $r_0 = 0.3$, $\Delta V_0 = +0.4 \text{ V}$), 43 % above the shipped envelope. This is a gap in the *shipped* design's validated envelope, exposed by this work rather than created by it; those corners are carried here as stability-only corners.

2.3. Drive channel

The drive channel is a VESC transport delay and current-loop lag feeding a force constant and a first-order longitudinal mode:

$$G_{22}(s) : \Delta i_{cmd} \rightarrow \text{Padé}_2(e^{-T_{d,v}s}) \rightarrow \frac{1}{\tau_v s + 1} \rightarrow \frac{k_t \eta_{dt} \varphi}{r_t} [\text{N A}^{-1}] \rightarrow \frac{1}{m_{eff}s + b_{eff}} \rightarrow K_{enc} K_v, \quad (5)$$

with DC gain $G_{22}(0) = K_{enc} K_v k_t \eta_{dt} \varphi / (r_t b_{eff}) = 3.7085 \text{ m s}^{-1} \text{ A}^{-1}$ and a mechanical pole at $-b_{eff}/m_{eff} = -0.1219 \text{ rad s}^{-1}$ — a near-integrator, an 8 s time constant. This recovers the powertrain form of [4] exactly when b is written shaft-referred; here it is written wheel-referred, with the motor's spinning loss reflected in through $b_{motor}(\varphi/r_t)^2$.

Two features of b_{eff} are counter-intuitive. At the nominal operating point aero contributes $0.0245 \text{ N s m}^{-1}$ and the motor's own free-run loss 0.3352, so the motor dominates the small-signal damping by $14\times$. And rolling resistance is Coulomb: it sets the operating-point torque (hence the coupling gains below) but its derivative with respect to v is zero away from standstill, so it must *not* appear in b_{eff} . The gain uncertainty $K_v \in \{0.5, 1, 2\}$ is structural, not a tolerance: the encoder sits downstream of the 9.49:1 reduction *and* the limited-slip differentials, so there is no fixed rotor-to-ground-speed mapping, and k_t is uncalibrated because no motor has been selected.

2.4. Coupling

$G_{21} = 0$ is a modelling *result*, not an assumption. The droop ratio reaches no state that feeds the speed output in either the design plant or the truth model, because the VESC runs a current loop: at fixed i_{cmd} the delivered torque is independent of v_{bus} until the duty rail. The residual this discards is measured — $|\Delta v_{bus}/\Delta r|$ at DC is 0.2002 V per unit r , 0.44 % of the 15.9 V bus over the full r span — nowhere near duty saturation anywhere in the envelope. It would break under deep battery discharge with a large i_{cmd} , or a regen event pushing the bus into overvoltage protection; both lie outside the linear envelope and are handled by the firmware's sequencing layer.

The live coupling is G_{12} , drive to share, in two stages. Linearizing motor input power $P = (k_t \omega i_m + R_m i_m^2)/\eta_v$ about the operating point at fixed V_{bus0} gives the bus-current draw

$$\Delta I_{bus} = A_i \Delta i_m + A_\omega \Delta \omega, \quad A_i = \frac{k_t \omega_0 + 2R_m i_{m0}}{\eta_v V_{bus0}}, \quad A_\omega = \frac{k_t i_{m0}}{\eta_v V_{bus0}}, \quad (6)$$

numerically $A_i = 0.24292 \text{ A A}^{-1}$ and $A_\omega = 3.9272 \times 10^{-4} \text{ A rad}^{-1} \text{ s}^{-1}$ at nominal. The $\Delta \omega$ term is what makes the coupling *dynamic* rather than a static feedthrough: it routes through the vehicle mode, so the coupling inherits the near-integrator pole. Differentiating (2) with respect to total current gives the share sensitivity

$$\frac{\partial \alpha}{\partial I_{tot}} = - \frac{\Delta V_0 r_0 (1 - r_0)}{k_d I_{tot0}^2}, \quad (7)$$

so that $G_{12}(s) = (\partial\alpha/\partial I_{tot})[A_i W_v(s) + A_\omega(\varphi/r_t)M(s)W_v(s)]/(\tau_f s + 1)$ with W_v the VESC path and M the vehicle mode. At nominal $\partial\alpha/\partial I_{tot} = -4.1667 \times 10^{-2}$ share per ampere and $G_{12}(0) = -2.7573 \times 10^{-2}$ share per ampere of i_{cmd} .

Both G_{11} and G_{12} pass through the *same* instance of the prefilter $1/(\tau_f s + 1)$, because there is exactly one such filter in firmware, downstream of the share estimate; the plant is therefore assembled as explicit block state-space rather than a block-diagonal composition, since duplicating the filter would add a spurious state and misrepresent the hardware. The nominal assembly is 8 states.

Sign uncertainty — the central limitation of the whole study.

Equation (7) is *exactly zero* at $\Delta V_0 = 0$ (matched sources: no coupling) and *flips sign* over the ± 0.4 V budget, which is an uncalibrated quantity. A centralized controller synthesized at one sign of ΔV_0 implements a coupling feedforward of the wrong sign at the mirror corner. Both signs are therefore mandatory in every evaluation below, and either outcome is a reportable result rather than a failure.

2.5. Why the RGA is the wrong instrument

The design plant is upper triangular, so its relative gain array is the identity — a gate confirms $\max |RGA - I| = 2.23 \times 10^{-16}$ across all frequencies and all cases, including $\Delta V_0 = \pm 0.4$. This is a property of *triangularity*, not of weak coupling: $RGA(T) = I$ for any triangular T no matter how large the off-diagonal entry. The RGA is blind to one-way coupling and must not be used as the coupling metric here. This corrects the instrument named in the outlook section of [8], whose prediction “RGA = I nominal” is confirmed — but confirmed *structurally*, so it carries no information about coupling strength.

The informative measures, computed on the scaled plant, are the directional ratio $|G_{s,12}|/|G_{s,11}|$ and the condition number (Table 2, Figure 1). At the nominal 2 A load the drive-to-share path is 39 % of the direct path; at 0.5 A it is $6.3\times$ it, because (7) scales as $1/I_{tot}^2$ — a full-authority motor transient can then move the share by more than a full-authority droop command can. That, not the RGA, is the number that motivates a centralized design.

Table 2. Coupling quantification on the scaled plant. The in-band window is DC to 200 rad s^{-1} : $\text{cond}(G_s(j\omega)) \rightarrow \infty$ as $\omega \rightarrow \infty$ for any strictly proper plant whose channels roll off at different orders, so a full-sweep maximum would measure roll-off asymmetry rather than coupling.

Quantity	Nominal (2 A, $r = 0.5$)	Light load (0.5 A)	FC cruise ($r = 0.85$)
peak $ G_{s,12} / G_{s,11} $	0.3928	6.2841	0.2613
$\text{cond}(G_s)$ at DC	5.3105	12.8707	6.8973
$\text{cond}(G_s)$, in-band	314.4	1896.3	238.4
$\partial\alpha/\partial I_{tot}$ [share/A]	−0.04167	− 0.66667	−0.02125
<i>Over the whole feasible operating grid (26 of 30 points feasible):</i>			
worst $ \partial\alpha/\partial I_{tot} $	1.120 share/A at 0.5 A, $r_0 = 0.7$, $\Delta V_0 = -0.4$ V		
worst DC $\text{cond}(G_s)$	13.08		
worst in-band $\text{cond}(G_s)$	2805.8		
$\max RGA - I $	2.23×10^{-16} — all cases, all frequencies		

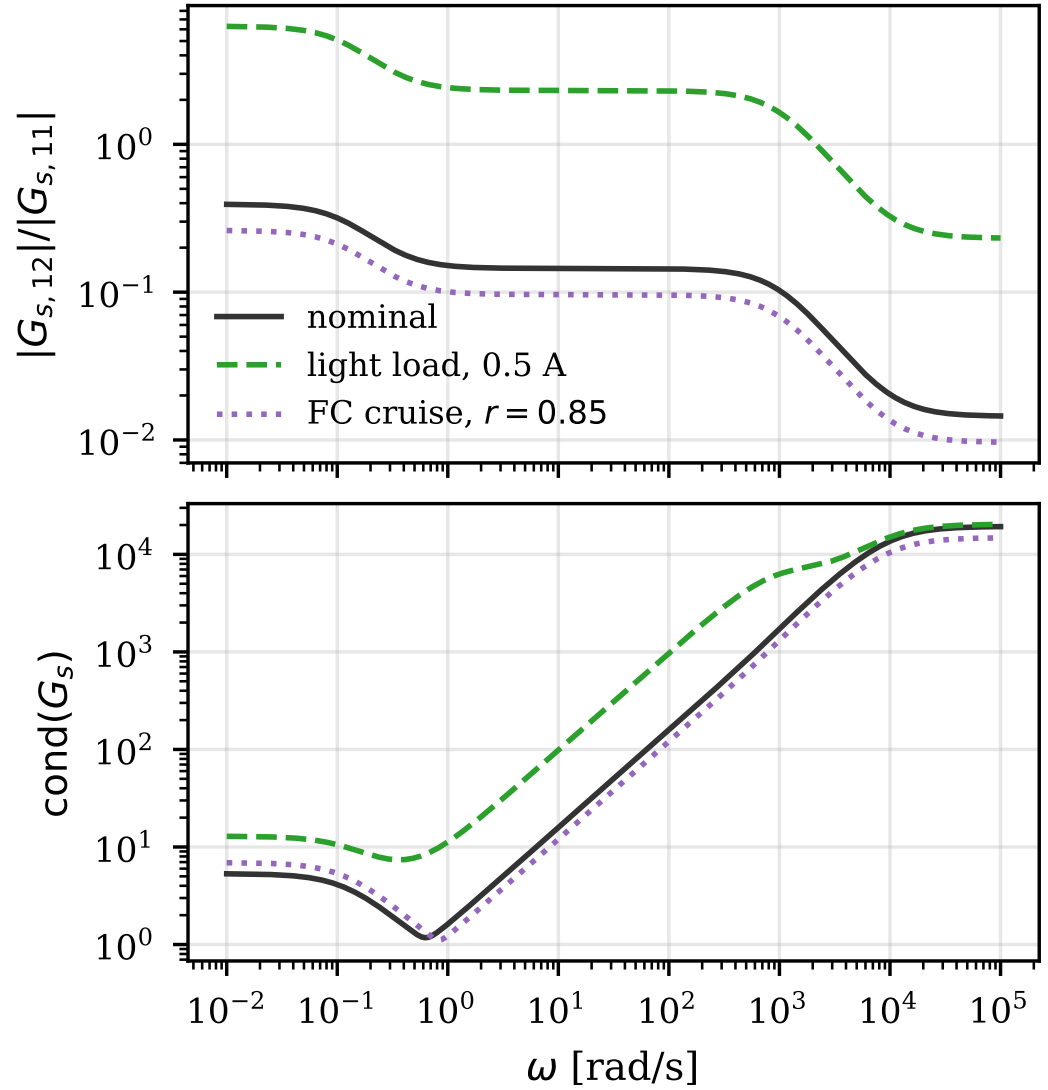


Figure 1. Coupling versus frequency for three scenarios (nominal 2 A $r = 0.5$; light load 0.5 A; FC cruise $r = 0.85$). Top: directional ratio $|G_{s,12}(j\omega)|/|G_{s,11}(j\omega)|$ on the scaled plant. Bottom: $\text{cond}(G_s(j\omega))$ for the same scenarios. The light-load curve exceeding unity across the band is the whole motivation for considering a centralized design; the RGA is identically I for all three and would show nothing.

2.6. Operating envelope, feasibility and corner families

An operating point is the 4-tuple $\text{OP} = (I_{\text{tot}0}, r_0, \Delta V_0, v_0)$, nominally $I_{\text{tot}0} = 2 \text{ A}$, $r_0 = 0.50$, $\Delta V_0 = +0.2 \text{ V}$ and $v_0 = 2 \text{ ms}^{-1}$. Half the $\pm 0.4 \text{ V}$ budget is used for ΔV_0 deliberately: synthesizing at zero would show the controller no coupling at all, and at ± 0.4 would over-commit to one sign.

The corner families are 10 operating points ($\{0.5, 2, 5\} \text{ A} \times \{0.3, 0.5, 0.7\}$ plus the FC-charge-cruise point $(2 \text{ A}, 0.85)$), 24 share corners ($\Delta V_0 \in \{-0.4, 0, +0.4\} \text{ V}$, $T_d \in \{0.5, 2\} \text{ ms}$, $\tau_r \in \{20, 300\} \mu\text{s}$, $\tau_f \in \{0, 0.8\} \text{ ms}$), and 24 drive corners ($K_v \in \{0.5, 1, 2\}$, damping factor $\in \{0.5, 2\}$, $\tau_v \in \{0.5, 5\} \text{ ms}$, $T_{d,v} \in \{1, 4\} \text{ ms}$). ΔV_0 lives on the uncertainty axis rather than the operating axis and overrides the operating point's value when the two are combined.

The full Tier-1 cross product is $10 \times 24 \times 24 = 5760$ corners, of which **768 are infeasible and are skipped, counted, and never silently linearized**: the RT1987 ideal-diode switches are unidirectional, so if (2) puts α_0 outside $(0, 1)$ the weaker source is blocked and the linearization is meaningless there. At 0.5 A the mismatch term reaches ± 0.56 share, so

(0.5 A, 0.5, ± 0.4) gives $\alpha_0 = 1.17$ or -0.17 . The remaining **4992 feasible corners** are the Tier-1 family used throughout.

Synthesis and all frequency-domain metrics run on the scaled plant $G_s = D_e^{-1} G D_u$ with $D_e = \text{diag}(0.05, 0.5)$ (maximum acceptable share and speed error) and $D_u = \text{diag}(0.35, 5.0)$ (half the r span, and the motor-current clamp). Nominally $G(0) = \begin{bmatrix} 1 & -0.02757 \\ 0 & 3.7085 \end{bmatrix}$, hence $G_s(0) = \begin{bmatrix} 7.0 & -2.757 \\ 0 & 37.09 \end{bmatrix}$ and $\text{cond}(G_s(0)) = 5.33$.

2.7. Validation against a full-order model

The design plant is checked against a 15-state converter-level truth model built by grafting the drive branch onto a verbatim copy (reproduced to 0.0 relative error) of the 11-state TPS61288 small-signal model used in the SISO work. That model is structurally matched — both channels share a reference and divider network — so ΔV_0 enters only through the realized current split: the droop conductances stay set by the commanded r_0 while the actual split becomes $\alpha_0 = r_0 + \Delta V_0 r_0 (1 - r_0) / (k_d I_{tot})$. That single change makes the coupling *emerge from the circuit*:

$$\delta\alpha = \frac{I_{B0} \delta i_F - I_{F0} \delta i_B}{I_{tot}^2}, \quad \delta i_F = -\frac{\delta v_{bus}}{k_d/r}, \quad \delta v_{bus} = -\delta I_{load} k_d \implies \frac{\delta\alpha}{\delta I_{load}} = -\frac{\Delta V_0 r(1-r)}{k_d I_{tot}^2}, \quad (8)$$

identical to (7). No coupling term is written into the truth model; the gate compares its DC $\partial\hat{\alpha}/\partial I_{load}$ against the design plant's analytic value and finds agreement to **0.032 %** on the odd-in- ΔV_0 part, over $I_{tot} \in \{0.5, 2, 5\} \text{ A} \times r_0 \in \{0.3, 0.5, 0.7\} \times \Delta V_0 \in \{0.2, 0.4\} \text{ V} \times C_{bus} \in \{30, 500\} \mu\text{F}$. A residual even-in- ΔV_0 offset of $\leq 2.4 \times 10^{-3}$ share per ampere is not a modelling error but a real asymmetry: the truth model's channels differ ($V_{in} = 9 \text{ V}$ vs. 8 V give different duty, compensation gain and right-half-plane zero) and the error amplifiers have finite DC gain, so a load step splits marginally unevenly even with matched references. Channel-by-channel frequency-response deviation over the design band is 0.93 % (share), 0.71 % (coupling) and 0.00 % (drive, shared construction) at nominal, worst 7.71 % over the full envelope against a 15 % gate.

3. The Two SISO Designs

3.1. Shipped share controller (prior work)

The share half of the decentralized baseline is the controller that ships in firmware today [8]: an H_∞ /Youla-H design on the SISO share plant (4) with $\gamma_{opt} = 0.6532$, built at a 5 % back-off ($\|T_{zw}\|_\infty = 0.6841$) and realized as three biquads plus a trapezoidal integrator at 1 kHz with back-calculation anti-windup. It achieves $T(0) = 1$ exactly by the scalar Youla-H correction, an S/T crossover near 110 rad s^{-1} , a nominal M_2 margin of 0.80 with 73.9° phase margin, and a 11.7 ms delay margin — roughly six times the worst modeled delay. Against a 60-corner family it is stable everywhere with worst-corner discrete $\|S\|_\infty = 1.87$, against 26.9 for the nominally tuned PI it replaced. It is used here *unchanged*.

3.2. New SISO drive controller

The drive half of the baseline did not exist and was built for this study, following the recipe of [4,5] on the drive plant (5). Its design record is worth reporting because it ran into a structural wall.

The papers' philosophy places a second-order strictly-proper weight on S , a makeweight roll-off weight on T , and a control-effort weight on Y . Two deviations were forced. First, the papers' W_d break at 18 rad s^{-1} sits *below* the W_p corner on this plant, making the two weights contend over the same decade, inflating γ and collapsing the achieved crossover; the break was moved to $2.5\times$ the W_p corner, preserving the intent (T

rolled off just above crossover). Second — the finding — the achieved crossover lands consistently *below* the W_p corner with $\gamma_{opt} \gg 1$ throughout: **the specifications are being shaped, not met**. W_p was therefore cornered at 50 rad s^{-1} to land the achieved crossover near the papers' 24 rad s^{-1} (Table 3).

Table 3. Drive-channel weight iteration (SISO baseline). W_d break fixed at $2.5\times$ the W_p corner throughout; $W_u = \text{makeweight}(dc, 300, hf)$. The chosen row is the most aggressive setting that still clears the 45° phase-margin gate and keeps worst-corner $\|S\|_\infty$ under the 2.5 target.

W_p corner [rad s ⁻¹]	W_u	γ_{opt}	ω_c [rad s ⁻¹]	PM [°]	worst $\ S\ _\infty$
24	(0.3, 300, 20)	9.14	12.3	56.6	1.64
24	(0.1, 300, 5)	5.60	15.2	46	—
40	(0.3, 300, 20)	13.25	16.9	54	1.89
50	(0.2, 300, 10)	13.03	20.75	48.9	2.41
60	(0.1, 300, 5)	—	24.4	42 (fails gate)	—

A γ of 13.03 means the weights are not met, not that the controller is bad: the delivered loop has $\|S\|_\infty = 1.3851$ nominal, 48.9° phase margin at 20.75 rad s^{-1} , 18.3 dB gain margin and a 41.12 ms delay margin, and is stable at all 72 drive corners with worst continuous $\|S\|_\infty = 2.4074$. The scalar Youla-H correction is tiny here ($T_H(0) = 0.99864$, a gain scale of 1.00137) and delivers $T(0) = 1.000000000000$.

The one implementation surprise is anti-windup, and it drives the cost comparison in Section 5. The reduced controller's *non-integral* branch alone has a DC gain of $367.7 \text{ A m}^{-1} \text{ s}^{-1}$, so it saturates the $\pm 5 \text{ A}$ actuator for speed errors above 13.6 mm s^{-1} and its lag states wind up independently of the integrator: the shipped integrator-only back-calculation scheme, applied here, produces a -0.298 m s^{-1} standing error and an 18.7 mm s^{-1} limit cycle on the 0 to 2 m s^{-1} step. The baseline therefore *requires* Hanus self-conditioning [6] on a five-state state-space realization, $x_{k+1} = (A_d - B_d C_d / D_d)x_k + B_d u / D_d$ with u the clamped output — exactly equivalent to the biquad-plus-integrator form when unsaturated (gated to $6.32 \times 10^{-10} \text{ A}$), but *not* implementable as a biquad cascade.

4. Centralized 2×2 H_∞ / Youla-H Synthesis

4.1. Weights and the DGKF solution

Synthesis uses block-diagonal weights on the $S/KS(Y)/T$ stack, one philosophy per channel. The share block is the shipped SISO set, unchanged. The drive block is *not* the Section 3 set — it was iterated independently against the MIMO corner battery — and the difference matters for how the two designs may be compared (Section 6.1). Table 4 gives the final set and the iteration that produced it.

Table 4. Final block-diagonal weights and the drive-block iteration. Bandwidth separation (share $\approx 110 \text{ rad s}^{-1}$ vs. drive 24 rad s^{-1}) is deliberate: the share loop rides on 1 ms electrical dynamics, the drive loop on an 8 s mechanical mode. All γ values are a-posteriori achieved levels.

Block	Share channel	Drive channel (final)
W_p (on S)	strictly_proper_lf($10^4, 40$)	strictly_proper_lf($10^4, 24$)
W_d (on T)	makeweight($0.5, 250, 40$)	makeweight($0.5, 60, 40$)
W_u (on Y)	makeweight($0.3, 600, 20$)	makeweight($0.5, 200, 20$)
<i>Iteration record (drive block):</i>		
it.1	papers' 2nd-order S weight	$\gamma = 2.2898$ — rejected, structurally unmeetable
it.2	1st-order family, same 24 rad s^{-1}	$\gamma = 1.4891$ — rejected, Tier-2 worst $\bar{\sigma}(S_o) = 2.17$
it.3a	relax W_u to $(0.15, 600, 20)$	$\gamma = 1.1580$ — <i>better</i> γ , rejected anyway (Tier-2 worst 2.41)
it.3b	W_u DC $0.3 \rightarrow 0.5$, break $300 \rightarrow 200 \text{ rad s}^{-1}$	$\gamma = 1.8144$ — final
share channel alone / drive channel alone / MIMO		0.5760 / 1.8144 / 1.8168

Iteration 1 is a finding about the papers' recipe, not a defect in it. A second-order strictly-proper S weight with corner $a = \omega_c / \sqrt{dc - 1}$ demands $|S| \sim s^2$ roll-in below a , i.e. *two* integrators in the loop. The drive plant's mechanical pole sits at 0.122 rad s^{-1} , so below that frequency the plant contributes no integration and only the controller's single integrator is available; the weight requires an integral crossover around 1200 rad s^{-1} to serve a 24 rad s^{-1} loop, a $50\times$ mismatch. Sweeping the W_p DC gain from 10^4 down to 4, the W_d break from 60 to 250 rad s^{-1} and the W_u DC from 0.3 to 0.04 never brought γ below about 1.3, with $\|W_p S\|$ binding at the low-frequency end. The papers' powertrain plant is a first-order lag whose pole is well above the weight corner, so the double-slope demand is served there; on SC001, with an 8 s mechanical time constant, it is not. The Section 3 SISO drive design reached the same conclusion by a different route.

Iteration 3a is the counter-intuitive one and is recorded as a lesson: a *lower* γ bought a *more aggressive* controller whose worst-corner sensitivity was worse. The binding constraint on this plant is corner robustness, not nominal performance, so γ **is the wrong scalar to minimize here**. Iteration 3b deliberately accepts a worse γ for corner margin, paying about 60 ms of small-signal drive settling. A methodological note belongs with it: an earlier pass appeared to clear the corner gate at a lighter W_u , but only because balanced truncation was being selected at a loose tolerance and the truncation error happened to damp the worst corner. Tightening the tolerance to 10^{-2} made the shipped controller a faithful reduction and the corner result a property of the design rather than of a lucky truncation.

The augmented plant has 14 states with $p_z = 6$, $p_y = 2$, $m_u = 2$. Two features of the solution deserve reporting.

The Y-Riccati degeneracy is a free cross-check.

With measurement-equals-reference structure $D_{21} = I_2$, so $Q_y = B_1(I - D_{21}^\top D_{21})B_1^\top = 0$ and $A_y = A - B_1 C_2$. In the augmented plant B_1 injects only into the W_p block row, so $A - B_1 C_2$ cancels that coupling exactly and is block-triangular with stable diagonal blocks: $Y = 0$ is *provably* the stabilizing solution, and a correct two-Riccati implementation must return it. Ours does, to 3.8×10^{-15} , validating the estimation half of the DGKF code path on a problem that does not otherwise exercise it.

The Riccati γ is optimistic in MIMO, and must not be gated on.

Because $Y \equiv 0$, the DGKF feasibility test $\rho(XY) < \gamma^2$ is vacuous, leaving only “X-Riccati solvable and positive semi-definite” — necessary but here not sufficient. The bisection reports $\gamma = 1.19173$; the central controller attains $\gamma_{achieved} = \mathbf{1.81684}$, with $\|T_{zw}\|_\infty = 1.81695 \leq 1.005\gamma$. In the 1×1 sub-problems the bisection is exact to four decimals, so the gap is a MIMO-only artefact of the degeneracy. A refined back-off bisection on the a-posteriori criterion is used so the delivered level is tight, and every γ quoted in this report is an a-posteriori achieved level. Independently, the MATLAB `hinfsyn` cross-check returns $\gamma = 1.8276$, between the bisection value and the shipped a-posteriori level, exactly as the degeneracy predicts.

Two structural gates follow for free. Because $G_{21} = 0$, any MIMO controller restricted to $e_1 = 0$ acts as a drive-channel SISO controller, so $\gamma_{MIMO} \geq \gamma_{drive}$ necessarily; measured, $1.8168 \geq 1.8144$, equal to 0.16 %. That confirms the machinery is consistent (a violation would prove a formula error) and shows **the coupling costs essentially nothing**: the centralized controller attains the drive channel’s own unconstrained optimum while also handling G_{12} . And at $\Delta V_0 = 0$ the plant is exactly diagonal, so the problem must decouple and $\gamma_{MIMO} = \max(\gamma_{share}, \gamma_{drive})$; measured, 1.814 13 vs. 1.814 43 — 0.02 % on a case with a known closed-form answer.

4.2. The MIMO Youla-H DC correction: $T(0) = I$

This is the report’s control-theoretic contribution. For a SISO plant G_P , the recipe of [5] takes the H_∞ -synthesized Youla parameter in zero-pole-gain form,

$$Y_H(s) = K_H \frac{\prod_{i=1}^n (s + z_i)}{\prod_{j=1}^m (s + p_j)}, \quad (9)$$

keeps every z_i and p_j , and rescales the gain alone to $K_{YH} = K_H / (Y_H(0)G_P(0))$, which enforces $T_{YH}(0) = G_P(0)Y_{YH}(0) = 1$ and, since $G_C = Y/S$ and $S(0) = 0$, plants at least one pure integrator in the realized controller.

A scalar gain has no matrix analogue that preserves “the same zeros and poles.” But it has one that preserves the *dynamics*: a constant *right* multiplier. Build the Youla parameter by state-space interconnection, $Y_H = K_H(I + G_s K_H)^{-1}$, valid because $D_K = 0$, so that $T = G_s Y_H$ exactly as in the scalar case. Then define

$$M := [G_s(0) Y_H(0)]^{-1} \in \mathbb{R}^{2 \times 2}, \quad Y_{YH}(s) := Y_H(s) M. \quad (10)$$

Since M multiplies only the input matrix of the state-space realization, the poles and zero dynamics of Y_H are untouched — the exact matrix analogue of “keep z_i and p_j , change K .” The correction is immediate:

$$T_{YH}(0) = G_s(0) Y_{YH}(0) = G_s(0) Y_H(0) [G_s(0) Y_H(0)]^{-1} = I. \quad (11)$$

The scalar recipe is recovered exactly when the problem is 1×1 : $M = 1 / (G_P(0) Y_H(0))$, i.e. $K_{YH} = K_H / (Y_H(0) G_P(0))$.

The choice of *right* multiplication is forced twice over. Algebraically, $T = G_s Y$, so only a right factor of Y can be inverted against $G_s(0) Y_H(0)$ to give the identity; $M Y_H$ would give $G_s(0) M Y_H(0) \neq I$ in general because matrices do not commute. Physically, left-multiplying would mix the controller *outputs* rather than re-map its inputs, changing which actuator answers which error — a different controller, not a corrected one.

The realization follows the scalar case block-for-block: $G_{c,YH} = Y_{YH}(I - G_s Y_{YH})^{-1}$, a positive feedback of G_s around Y_{YH} , well-posed because $D_Y = 0$. The final step is

what makes the property robust rather than merely achieved: an ordered real Schur decomposition plus Sylvester block-diagonalization separates the two near-origin poles of $G_{c,YH}$ into an exact residue bank K_I/s plus a stable remainder. **After the split**, $T(0) = I$ **is structural** — two exact integrators with a non-singular K_I force $S(0) = 0$ regardless of any subsequent rounding, truncation or float32 quantization, so the DC property no longer depends on a numerically delicate pole-at-almost-zero, which is the risk the scalar recipe carries into a fixed-point implementation. Numerically,

$$K_I = \begin{bmatrix} 7.885807 & 0.1429398 \\ 3.189967 \times 10^{-4} & 0.3705096 \end{bmatrix}, \quad (12)$$

nearly lower triangular with a small (1,2) entry and a negligible (2,1) entry: the DC coupling correction runs drive $\rightarrow$ share, matching the plant's triangularity.

The correction is a nudge, not a rescue, and that is the healthy outcome. Here $\text{cond}(G_s(0)Y_H(0)) = 1.0001$ and $\|M - I\|_2 = 1.75 \times 10^{-4}$, saying the H_∞ controller was already integral to four decimals, as one expects from an S weight with DC gain 10^4 (a large $\|M - I\|$ would have indicated the weights, not the realization, needed work). The delivered $\|T(0) - I\|_{\max}$ is 7.3×10^{-10} , preserved through reduction. The correction's remaining job is therefore not to *create* the integrator — H_∞ already put a near-integrator there — but to fix its *direction*.

4.3. Is the centralized controller actually using the coupling?

Table 5 answers this, and the answer is the report's headline.

Table 5. Peak scaled magnitude of each controller entry, and its ratio to the diagonal term of the same row.

Entry	Peak $ \cdot $ (scaled)	Ratio to its diagonal
K_{11} (share $\leftarrow$ share error)	78.86	—
K_{12} (share $\leftarrow$ speed error)	1.4955	1.90 %
K_{21} (drive $\leftarrow$ share error)	4.09×10^{-3}	0.085 %
K_{22} (drive $\leftarrow$ speed error)	4.792	—

$K_{21} \approx 0$ is correct and expected: $G_{21} = 0$, so there is nothing the drive actuator can do about a share error, and its 8.5×10^{-4} ratio is numerical residue — a positive check that the synthesis discovered the plant's triangularity by itself. K_{12} at 1.9 % of K_{11} is the real coupling feedforward: the share actuator pre-empts the bus-current disturbance a commanded acceleration is about to create. It is small in *ratio* because K_{11} must be large (the share loop's 110 rad s^{-1} bandwidth), not because the term is inactive. Its magnitude is bounded by the ΔV_0 sign uncertainty by construction: a feedforward built at $+0.2 \text{ V}$ is the wrong sign at -0.4 V , and the synthesis correctly declines to bet much on it.

The centralized controller is about 98 % decentralized by gain. That is the sharpest available statement that decentralized control here is justified, not merely convenient — and it comes from the centralized design itself.

5. Implementation

The controller $G_{c,YH}$ has order 30 (14 controller + 8 plant + 8 from the Youla inter-connection); after the integrator split the stable remainder is order 28. Balanced truncation selects order 5 at a 10^{-2} margin, with a maximum relative $\bar{\sigma}$ deviation of 4.07×10^{-3} over $\omega \in [0.1, 10^4]$ against a 2×10^{-2} gate. Deviation is measured per frequency rather than against a single global scale, which would be dominated by the integrator's low-frequency

magnitude and would make the gate vacuous. The final continuous controller is **7 states**: two exact integrators plus a five-state remainder.

The whole controller runs single-rate at 2 ms (500 Hz), because it issues the motor current command and the VESC UART frame floor caps that channel there. The cost to the share channel is bounded and checked, not assumed: Nyquist is about 1571 rad s^{-1} , still $14\times$ the 110 rad s^{-1} share crossover, and the extra effective delay is at most 1 ms — inside the shipped SISO design's 11.7 ms delay margin and covered by the $T_d = 2 \text{ ms}$ corner. The entire Tier-1 family is re-verified in discrete time (0 of 4992 unstable, worst $|z| = 0.999761$, which is the exact integrator at the sample rate, not a marginal design pole).

The integrator is an exact 2×2 Tustin block with the matrix K_I , kept separate from the remainder so anti-windup can act on it; the recursive trapezoid form is gated equal to the LTI form at 8.3×10^{-17} . *Ordering matters*: the integrator is advanced first and the new value used in the output, matching the shipped firmware idiom — advancing it afterwards inserts a hidden extra sample of delay, which the equivalence gate caught. The stable remainder is discretized by Tustin and transformed to real modal (block-diagonal) form, $A_{\text{modal}} = \text{diag}(-0.9876, 0.7558, 0.6290, \begin{bmatrix} -0.0736 & 0.2009 \\ -0.2009 & -0.0736 \end{bmatrix})$, gated at $\text{cond}(T) = 2.97$ and reproducing the Tustin response to 9.2×10^{-16} . The rationale is float32: a modal A has the pole locations themselves as its coefficients, so rounding perturbs each mode independently instead of perturbing companion-form polynomial coefficients, where a 10^{-7} relative error can move clustered roots by far more.

Anti-windup has two parts. Because the integrator state adds directly to the scaled input, its output map is I_2 and the back-calculation map is exactly D_u^{-1} : on saturation, $x_{\text{int}} += D_u^{-1}\delta$ with $\delta = u_{\text{sat}} - u$ physical, and $\text{cond}(D_u^{-1}) = 14.29$ so no diagonal fallback is needed — the directionality-preserving generalization of the shipped scalar scheme. On top of it, x_{int} is clamped componentwise to $[-1, 1]^2$: the integrator alone may never demand more than the actuator can deliver. The second term is needed here and not in the shipped SISO controller because bare back-calculation dumps the *whole* clamp excess into the integrator, including the part created by the direct feedthrough. The shipped scalar controller's direct term is small, so this is harmless there; here $D_{22} \approx 0.45$ scaled against a peak $|K_{22}|$ of 4.8, so bare back-calculation let the integrator absorb a direct-term excess and then hold the *opposite* rail once the error collapsed. It was caught by a closed-loop recovery simulation (rail at -5 A , hold 2 s, release: rail fraction 1.00 during the hold, immediate release, $1.4 \times 10^{-4} \text{ m s}^{-1}$ recovery tail, no sustained rail). An open-loop “hold a huge error then zero it” test cannot reveal it, because a loop with an externally pinned error has no mechanism to unwind.

The drive channel is actuator-limited, and every dynamic claim must say so.

With $K_F = k_t \eta_{dt} \varphi / r_t = 1.334 \text{ N A}^{-1}$ and $m_{\text{eff}} = 2.95 \text{ kg}$, the $\pm 5 \text{ A}$ clamp bounds acceleration at 2.26 m s^{-2} . A 0.5 m s^{-1} reference step cannot be tracked at the 24 rad s^{-1} design bandwidth — it takes about 0.22 s just to accelerate. The share step (0.05 share) settles to 2 % in 44 ms with final α exact; the small-signal drive step (0.05 m s^{-1} , peak 1.3 A) settles in 248 ms with final v exact; the actuator-limited drive step (0.5 m s^{-1} , peak 5.0 A) takes 23.2 s and lands at 0.499 887 — an actuator-limited slew plus a slow integral tail (post-saturation recovery is governed by $K_{I,22} \approx 0.37$ against a 0.122 rad s^{-1} plant pole), *not* a windup hang; the gate proves the $T(0) = I$ property survives a saturation episode unbiased. The 24 rad s^{-1} bandwidth is a small-signal claim, valid for speed excursions below roughly 0.1 m s^{-1} .

Cost.

Table 6 gives the Teensy 4.1 comparison, and the result is counter-intuitive: **the centralized controller is the cheaper implementation.**

Table 6. Implementation cost on the Teensy 4.1 (600 MHz Cortex-M7). The decentralized state count is quoted as implemented; its drive half needs a six-state Hanus self-conditioned realization rather than the five-state controller order, because the conditioning product adds a state's worth of storage and an n^2 term per tick.

Quantity	Decentralized	MIMO
rate	1 kHz (share) + 500 Hz (drive)	500 Hz single-rate
controller states (implemented)	$4 + 6 = 10$	7
MAC per tick	18 @ 1 kHz, 63 @ 500 Hz	57 @ 500 Hz
MAC per second	49 500	28 500
ratio MIMO / dec	—	0.576
coefficient floats	—	65
float32 replay error (64-step, saturating)	—	1.79×10^{-6} (gate 5×10^{-4})

The explanation is the anti-windup finding of Section 3: the decentralized baseline *cannot* be implemented as a biquad cascade. Its drive half needs the Hanus five-state state-space form, whose $A_d - B_d C_d / D_d$ product costs an extra n^2 per tick, and it runs one of its two halves at 1 kHz. The single-rate centralized controller is 7 states with a 5×5 modal A of which only 7 entries are non-zero beyond round-off.

Both figures are negligible on this core (under 0.1 % of a single issue slot), so **the compute cost is not the cost.** The genuine costs of centralization are: it moves the share loop from 1 kHz to 500 Hz and couples the two control tasks into one scheduling unit; float32 coefficient sensitivity (measured, mitigated by the modal form and by keeping D_e / D_u out of the matrices); and a matrix-valued anti-windup scheme that is more code to get right than the shipped scalar one. The UART frame floor is unchanged either way and remains the true bottleneck.

6. Comparison

6.1. Setup and what makes it fair

Three controllers are compared. dec is the decentralized baseline, blkdiag(K_{share}, K_{drive}), with the shipped share controller at 1 kHz and the Section 3 drive controller at 500 Hz. mimo is the Section 4 controller at 500 Hz. dec500 is the decentralized pair with its share half re-discretized at 500 Hz, and exists purely to isolate the sample-rate difference as a confound.

The essential fairness condition is that **both controllers are closed against the same coupled 2×2 plant**, using the same assembly routine, corner sets and simulations; the decentralized controller is never evaluated on a diagonal plant, which would flatter it by construction. Time-domain results use a multirate discrete simulator at a 0.1 ms base step with the plant ZOH-discretized per corner and controller outputs held between ticks, so dec runs its two halves at their native rates. Frequency-domain results use continuous physical-coordinate controllers recovered from the emitted float32 headers by exact inverse Tustin plus the analytic K_I/s integrator; this validates the emitted artefacts at the cost of about 10^{-7} relative coefficient truncation, far below the reported precision. All time-domain signals are small-signal deviations about the nominal operating point; every step amplitude is stated and every actuator-limited case flagged.

One limitation must be stated rather than buried: the two designs use different weight families on the drive channel (second-order at 50 rad s^{-1} for the SISO baseline, first-order at 24 rad s^{-1} for the MIMO) and their γ values sit on different plant scalings, so γ is **not comparable between them and is never quoted side by side**. An apples-to-apples weight set would have required coordinating the two syntheses; the honest question — which controller performs better on the real coupled plant? — is answered by the closed-loop harness regardless.

6.2. Stability across the Tier-1 family

Table 7. Tier-1 stability, 4992 feasible corners. The discrete check for dec is not an approximation: it is the exact two-step monodromy of the 1 ms-lifted periodic loop.

Controller	cont. unstable	worst $\text{Re}(\lambda)$ [rad s ⁻¹]	disc. unstable	worst $ z $
dec (multirate)	0 / 4992	−0.00400	0 / 4992	0.999992
dec500	—	—	0 / 4992	0.999992
mimo (single-rate)	0 / 4992	−0.11976	0 / 4992	0.999761

No decentralized instability was found anywhere in the Tier-1 family, in continuous time or in the exact multirate discrete loop — a positive result for the decentralized architecture, reported as such. The one asymmetry worth naming is that its slowest closed-loop mode is $0.0040 \text{ rad s}^{-1}$ (a 250 s time constant) against the centralized loop's $0.1198 \text{ rad s}^{-1}$, a factor of 30: both are stable, but the decentralized pair leaves a very slow, nearly-cancelled mode that the centralized design does not.

6.3. Metric 1 — worst-corner sensitivity

Table 8. Sensitivity at the nominal point and over the Tier-2 family (168 in-envelope corners of 208, the remainder in two documented waiver families). The waiver policy is inherited verbatim so both controllers are judged on identical terms.

Quantity	dec	mimo	verdict
<i>Nominal plant:</i>			
$\ S_o\ _\infty$ ($\bar{\sigma}$ peak)	1.3974	1.2096	MIMO
$ S_{11} $ peak (share)	1.2523	1.2094	MIMO
$ S_{22} $ peak (drive)	1.3851	1.1079	MIMO
$\ T_{\alpha \leftarrow v_{ref}}\ _\infty$ [share/(m/s)], continuous	0.25676	0.00088	MIMO (293×)
same, on the implemented 2 ms loop	0.25676	0.00919	MIMO ($\sim 28\times$)
slowest closed-loop mode [rad s ^{−1}]	0.00400	0.1219	MIMO
<i>Tier-2, 168 in-envelope corners:</i>			
worst $\bar{\sigma}(S_o)$	2.5535	1.9153	MIMO, −25 %
worst $ S_{11} $ peak	1.5769	1.3695	MIMO
worst $ S_{22} $ peak	2.4062	1.3918	MIMO, −42 %
worst $\ T_{\alpha \leftarrow v_{ref}}\ _\infty$	0.8480	1.1467	dec
continuous-unstable corners	0	0	tie
corners where MIMO has lower $\bar{\sigma}(S_o)$	156 of 168		MIMO
<i>Waived families (stability-gated only):</i>			
worst $\bar{\sigma}(S_o)$, FC cruise ($r_0 = 0.85$)	2.4960	1.8203	MIMO
worst $\bar{\sigma}(S_o)$, K-out-of-envelope	5.7702	10.8242	dec

Two corrections to how these numbers must be quoted.

Both come from the independent MATLAB cross-check (Section 7) and both were subsequently reproduced exactly in Python; they supersede the framing used in earlier drafts of the design record.

First, **the Tier-2 worst $\bar{\sigma}(S_o) = 1.9153$ is a subsample lower bound, not the full-grid worst.** A full 24×24 sweep at the nominal operating point found $\bar{\sigma}(S_o) = 2.0345$ at a corner not in the representative set ($\Delta V_0 = -0.4$, $T_d = 2$ ms, $\tau_r = 300$ μ s, $\tau_f = 0.8$ ms, $K_v = 2$, damping factor 2, $\tau_v = 0.5$ ms, $T_{d,v} = 4$ ms). Stability is unaffected (576/576 in that sweep, and Tier-1 remains 0/4992). The consequence is that the “< 2.0” performance gate number is subsample-dependent and the true full-grid worst is at least 2.03. The decentralized comparison point (2.5535) comes from the same subsample, so the *relative* verdict — a 20 to 25 % reduction in worst-corner $\bar{\sigma}(S_o)$ — stands; the absolute gate claim does not.

Second, **the $293 \times$ nominal cross-coupling rejection is a continuous-domain figure.** On the as-implemented 2 ms ZOH loop, $\|T_{\alpha \leftarrow v_{ref}}\| = 9.19 \times 10^{-3}$ (MATLAB and Python agree to four digits), i.e. about $28 \times$ better than decentralized, not $293 \times$. The time-domain excursion metrics below were always simulated at the implemented rates and are unchanged.

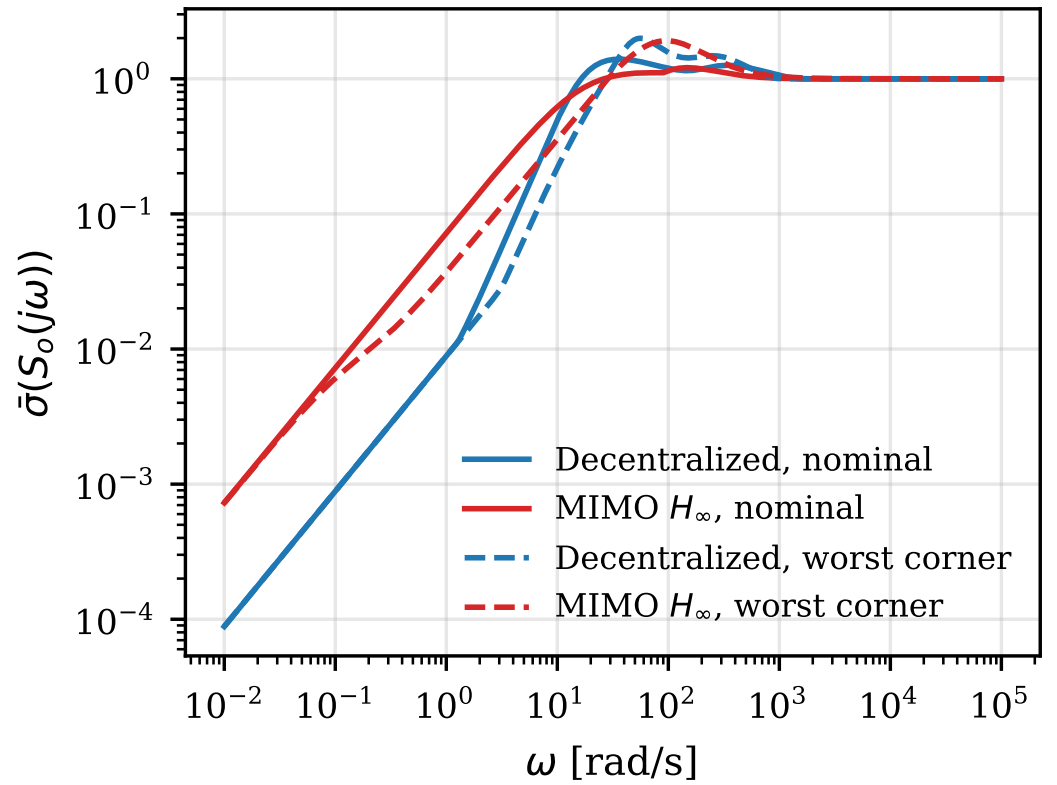


Figure 2. $\bar{\sigma}(S_o(j\omega))$ for both controllers, at the nominal operating point and at the worst in-envelope corner ($I_{tot0} = 2$ A, $r_0 = 0.5$, $\Delta V_0 = -0.4$ V, with the drive channel’s gain and delay extremes). Blue is decentralized, red is MIMO H_∞ .

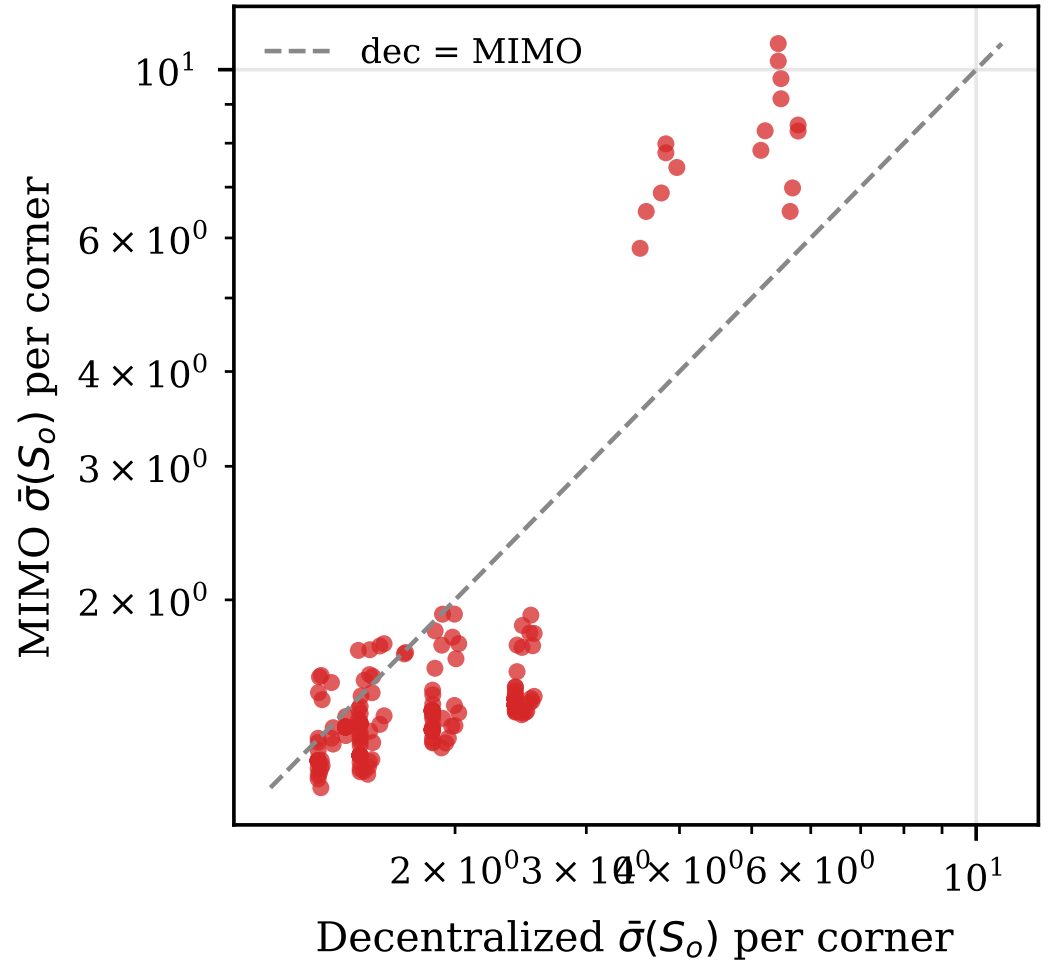


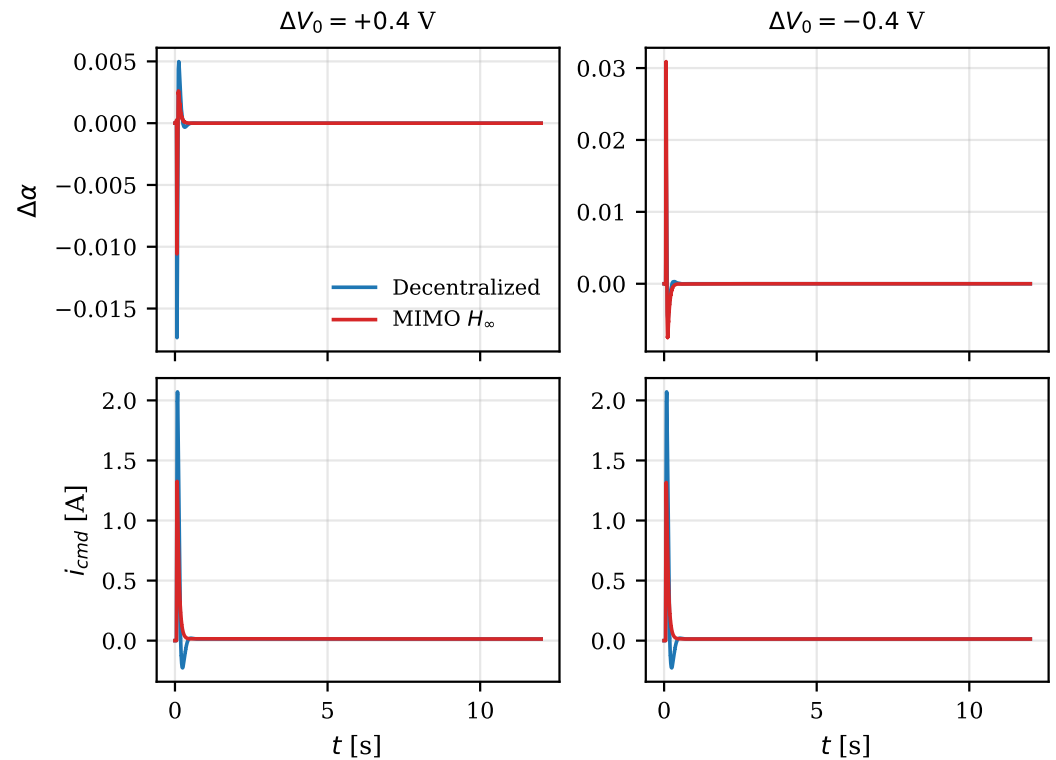
Figure 3. Per-corner $\bar{\sigma}(S_o)$, decentralized (horizontal) against MIMO (vertical), with the $y = x$ reference line. Points below the line favour the centralized controller: 156 of 168 in-envelope corners lie there. The out-of-envelope corners — light load with full mismatch, where the share plant gain reaches $K \approx 2.07$ — are the visible cluster in which the centralized controller is the worse of the two.

6.4. Metric 2 — drive-transient share excursion

A speed-reference step is applied at $t = 0.05$ s. The small step ($\Delta v_{ref} = +0.05 \text{ m s}^{-1}$) is fully linear — zero time on the current rail — and is the clean comparison. The large step ($\Delta v_{ref} = +2 \text{ m s}^{-1}$) puts the loop on the ± 5 A rail for 1.3 to 1.5 % of the horizon and is therefore *actuator-limited*: it measures a plant-and-actuator property and an anti-windup recovery, not a linear design.

Table 9. Drive-transient share excursion. Amplitudes are deviations about $v_0 = 2 \text{ m s}^{-1}$; horizons are 12 s (small) and 60 s (large).

Quantity	dec	dec500	mimo
<i>Small signal, $\Delta v_{ref} = +0.05 \text{ m s}^{-1}$ (0 % rail):</i>			
$\max \Delta\alpha , \Delta V_0 = +0.4$	0.01744	0.01868	0.01062
$\max \Delta\alpha , \Delta V_0 = -0.4$	0.01744	0.01868	0.03096
$\Delta\alpha$ settle (< 0.002) [s]	0.189	0.189	0.139
v settle (2 %) [s]	0.358	0.358	0.298
peak $ i_{cmd} $ [A]	2.071	2.071	1.326
MIMO/dec excursion ratio	—	—	0.609 (+) / 1.775 (-)
<i>Large signal, $\Delta v_{ref} = +2 \text{ m s}^{-1}$ (actuator-limited):</i>			
$\max \Delta\alpha , \Delta V_0 = +0.4$	0.0885	0.0903	0.2486
$\max \Delta\alpha , \Delta V_0 = -0.4$	0.0885	0.0903	0.4521
v settle (2 %) [s]	1.141	1.141	13.229
rail fraction	0.0149	0.0149	0.0131

**Figure 4.** Small-signal drive transient, $\Delta v_{ref} = +0.05 \text{ m s}^{-1}$ (zero time on the current rail). Left column $\Delta V_0 = +0.4 \text{ V}$, right column $\Delta V_0 = -0.4 \text{ V}$; top row share deviation, bottom row motor current command. The decentralized traces are identical between columns — as they must be, since a diagonal controller on a linear plant sees only the magnitude of the coupling, not its sign — while the centralized traces invert.

This is the cleanest single result in the study. At the sign of ΔV_0 the centralized controller was synthesized for, it cuts the drive-induced share excursion by 39 % and settles faster with a 36 % smaller current peak; at the mirrored sign it makes the excursion 78 % worse. Splitting the corner statistics by sign over the 96 coupled in-envelope corners ($\Delta V_0 \neq 0$; at $\Delta V_0 = 0$ both cross-transfers are zero to machine precision and the ratio is 0/0 noise), the median cross-transfer ratio MIMO/dec is **1.370** when the sign matches the

synthesis and **3.198** when it opposes, spread 0.3154 to 7.267; the centralized controller has the smaller cross-transfer at only 24 of 96 corners.

On the saturated large step the decentralized pair is clearly better — 2.8 to $5.1 \times$ smaller share excursion and an $11.6 \times$ faster speed settle. The mechanism is that of Section 5: the centralized drive integrator residue is $O(0.6)$ against a 0.122 rad s^{-1} vehicle pole, so post-saturation recovery is slow *by design*, and the authority clamp bounds but does not eliminate the transfer. The 60 s horizon was chosen so this is measured, not reported as a horizon artefact.

6.5. Metric 3 — regen event on the truth model

The regen event is run against the 15-state truth model with a bus-voltage output, not the 8-state design plant. Δv_{ref} is stepped to -2 m s^{-1} (2 m s^{-1} to standstill) and i_{cmd} rails at -5 A ; the 0.8 ms prefilter is applied digitally at the base rate since the truth model carries no τ_f by convention. Horizon 60 s.

Table 10. Regen event, 15-state truth model. The residual Δv_{bus} of 0.105 V is the new steady-state load operating point (the vehicle has stopped), not a tracking error; recovery is therefore measured about the final value.

Quantity	dec	dec500	mimo
$\max \Delta v_{bus} \text{ [V]}$	0.42279	0.42279	0.41578
$\Delta v_{bus} \text{ final [V]}$	0.10478	0.10478	0.10477
$v_{bus} \text{ recovery [s]}$	1.295	1.295	1.036
$\max \Delta \alpha $	0.04582	0.04761	0.29860
peak $i_{cmd} \text{ [A]}$	-5.000	-5.000	-5.000
$v \text{ settle (2 \%) [s]}$	1.141	1.141	13.229

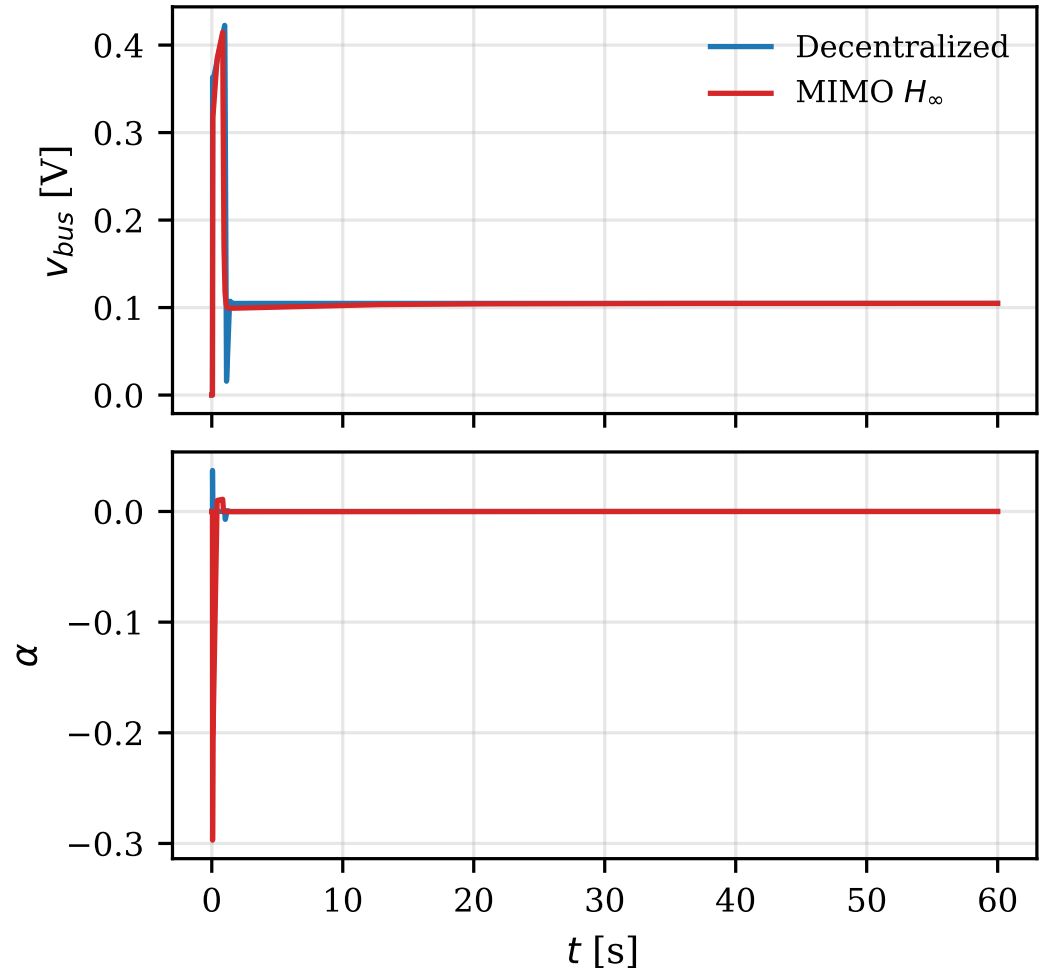


Figure 5. Regen event on the 15-state truth model. Top: bus-voltage deviation. Bottom: share deviation. The bus traces are nearly indistinguishable — during the saturated part of the event both controllers sit on the same -5 A rail, so the bus sees the same disturbance and neither droop loop has authority over it.

The bus excursion is essentially controller-independent: 0.4228 V against 0.4158 V, a ratio of 0.983. This is the physically expected answer and is worth stating explicitly, because the outlook section of [8] proposes this metric as a discriminator between the two architectures. **It is not one.** The centralized controller pays a $6.5\times$ larger share excursion for a 1.7 % smaller bus excursion.

6.6. Supporting cases: FC-charge cruise and the drive cycle

Two further scenarios were run because they probe regimes the headline metrics do not.

At the FC-charge-cruise operating point (2 A, $r_0 = 0.85$, static share $\alpha_0 = 0.8925$) the energy-management strategy parks the droop ratio on its upper clamp so the fuel cell carries the bus and the battery charges; the share loop has no upward authority left by construction. A $+0.5\text{ m s}^{-1}$ drive step over a 60 s horizon shows **no windup interaction for either controller**: both saturate r on the upper clamp for most of the horizon (0.997 and 0.798), both use only the small downward authority available ($r_{min} = 0.84434$ and 0.84987), both converge to a $\sim 2 \times 10^{-3}$ residual share error — the authority loss itself, not a hang — and neither shows a limit cycle (tail peak-to-peak $\leq 10^{-5}$). All three anti-windup schemes behave gracefully in the degenerate-authority regime, closing out a risk flagged at the start of this work. Peak share excursions are 0.026 97 (dec) and 0.026 28 (MIMO): a tie.

The 30 s drive-cycle profile (standstill; ramp 0 to 2 m s^{-1} over 5 s; cruise; coast to 0.5 m s^{-1} ; hold; stop) with a $\Delta\alpha_{ref} = +0.2$ share step over 11–15 s and a -1.0 A load disturbance over 22–25 s is dominated by the actuator-limited ramps, so whole-cycle RMS speed errors are similar (0.2026 vs. 0.2113 m s^{-1}). The *cruise window* discriminates, and there the decentralized pair is far better: numerical zero ($2.81 \times 10^{-7} \text{ m s}^{-1}$ RMS speed error) against 0.0633 m s^{-1} for the centralized controller, still recovering from ramp saturation. On the isolated load disturbance the two are equivalent to within 2 %. One result of note: the centralized controller drives the droop ratio to its *lower* clamp ($r = 0.15$) during the cycle, whereas the decentralized pair never goes below 0.44 — the cross-channel term spending droop authority in response to a saturated drive channel, a real cost of centralization on this plant.

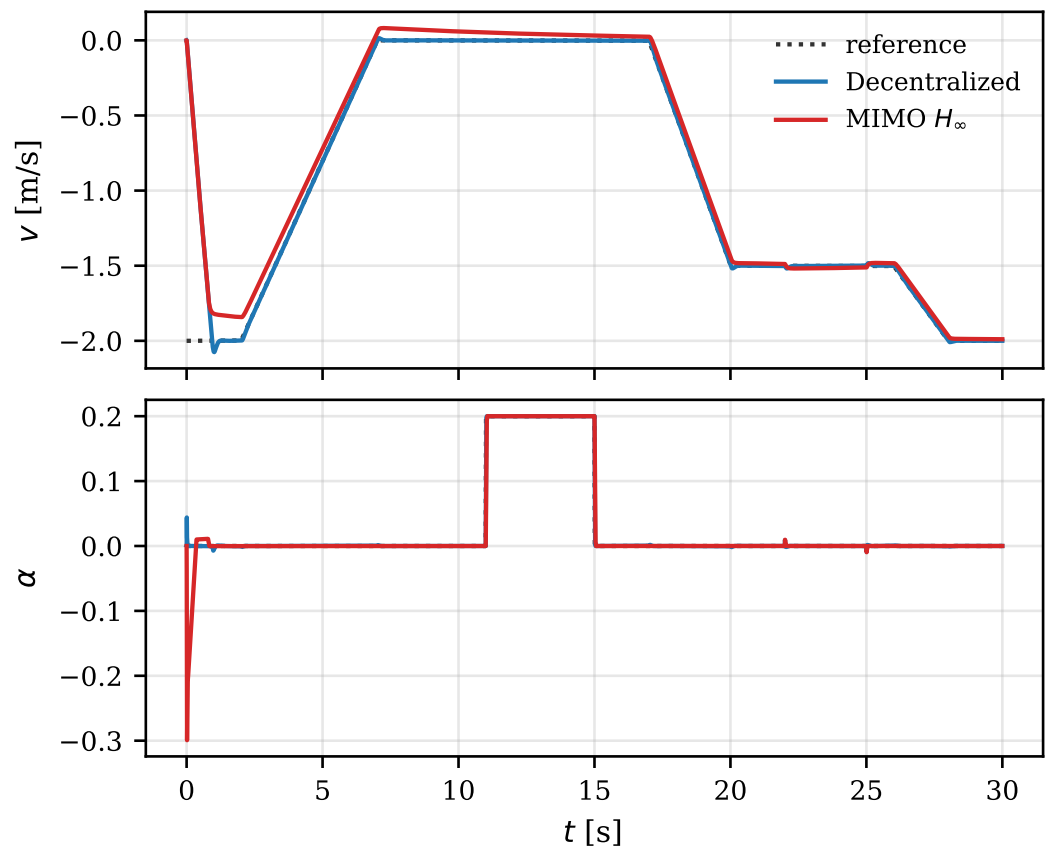


Figure 6. 30 s drive cycle. Top: speed against reference. Bottom: share against reference. The whole-cycle RMS errors are dominated by the actuator-limited ramps; the cruise segment (8–11 s) is where the two controllers actually differ.

6.7. The rate confound is isolated, and is not the explanation

dec500 — the decentralized pair with its share half re-discretized at 500 Hz — changes the share excursion by only 2.0 % on the large step and 7.1 % on the small step, leaves the regen bus excursion bit-identical (ratio 1.000), leaves Tier-1 stability at 0/4992, and changes the drive-cycle RMS errors in the fourth significant figure. **Every conclusion above survives equalizing the sample rates.**

6.8. The three headline metrics give three different answers

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Table 11. The three metrics named in the outlook section of [8], and their verdicts.

Metric	dec	mimo	Verdict
worst-corner $\ S\ _\infty$ (in-envelope $\bar{\sigma}(S_o)$)	2.5535	1.9153	MIMO wins, −25 % (both on the same subsample; full-grid worst is ≥ 2.03)
drive-transient share excursion, small signal, $\Delta V_0 = +0.4 / -0.4$	0.01744 / 0.01744	0.01062 / 0.03096	split by ΔV_0 sign
regen-event bus ex- cursion	0.422 79 V	0.415 78 V	tie (ratio 0.983)

That the three metrics disagree is the result. Worst-corner sensitivity favours the centralized design genuinely and robustly. The drive-transient excursion is decided by a sign the hardware has not yet pinned down — and the nominal-point $293\times$ (continuous) or $28\times$ (implemented) cross-transfer improvement is real but is a nominal-point number and must be quoted as such. The regen bus excursion discriminates nothing, because both controllers are on the same rail. Adding the operational metrics the outlook section does not name, the decentralized pair is substantially better on every saturated large-signal case.

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6.9. Caveats

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1. **ΔV_0 sign uncertainty bounds any feedforward benefit.** The entire centralized coupling advantage lives in (7), which is zero at $\Delta V_0 = 0$ and flips sign with it. Until ΔV_0 is measured, the honest claim is “MIMO buys coupling rejection at the design sign and costs it at the mirror sign.”
2. **The K -out-of-envelope corners are waived, and the centralized controller is worse there** — $\bar{\sigma}(S_o) = 10.82$ against 5.77. The stability-only waiver is physically justified: these are the 16 light-load, full-mismatch corners where $K \approx 2.1$ and the static share sits within ~ 0.02 of the unidirectional-switch feasibility boundary, so the plant model itself is at the edge of validity. The direction of the difference is a cost of centralization and is reported, not hidden.
3. **Large steps are actuator-limited.** The 2 m s^{-1} step and the drive-cycle ramps saturate the $\pm 5 \text{ A}$ clamp; those comparisons are as much about anti-windup recovery as about linear design.
4. **The Tier-2 worst-corner numbers are a subsample** (see above): the relative verdict survives, the absolute gate value does not.
5. **The shipped-share re-derivation carries a documented 0.67 % integrator-gain offset** (112.679 here against 111.930 in the shipped firmware header), a solver-accuracy propagation from the γ_{opt} anchor. It is well inside the corner family and affects no comparison above.
6. **Nothing here is bench-calibrated.** Every calibration item remains open — ΔV_0 above all, plus k_t , the encoder chain, m_{eff} , τ_v and $T_{d,v}$. The corner families are wide precisely because of this, but a calibrated plant could move the verdicts, particularly the second headline metric.

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7. Independent Validation

The entire chain was cross-checked in MATLAB (Control System and Robust Control Toolboxes) by a script that independently rebuilds the plant from the documented equations, re-runs `hinf syn` and the MIMO Youla-H correction, parses the emitted coefficient header, and re-runs the corner battery and the transients. **The verdict is PASS on all six criteria.**

Table 12. MATLAB cross-validation anchors.

Anchor	MATLAB	Python
design-plant DC gain (max abs. deviation)		2.2×10^{-9}
<code>hinf syn</code> γ	1.8276	1.1917 (bisection) / 1.8168 (a-posteriori)
$\ M - I\ _2$	1.758×10^{-4}	1.75×10^{-4}
$\text{cond}(G_s(0)Y_H(0))$	1.0001	1.0001
$T(0)$ on the parsed header		I to 10^{-9}
a-posteriori $\ T_{zw}\ _\infty$ of the shipped artefact	1.8175	1.8168 (0.04 %)
nominal $\bar{\sigma}(S_o)$	1.2596	1.2095
corner battery		576 / 576 stable
share excursion, $\Delta V_0 = +0.4 / -0.4$	0.01023 / 0.03037	0.01062 / 0.03096

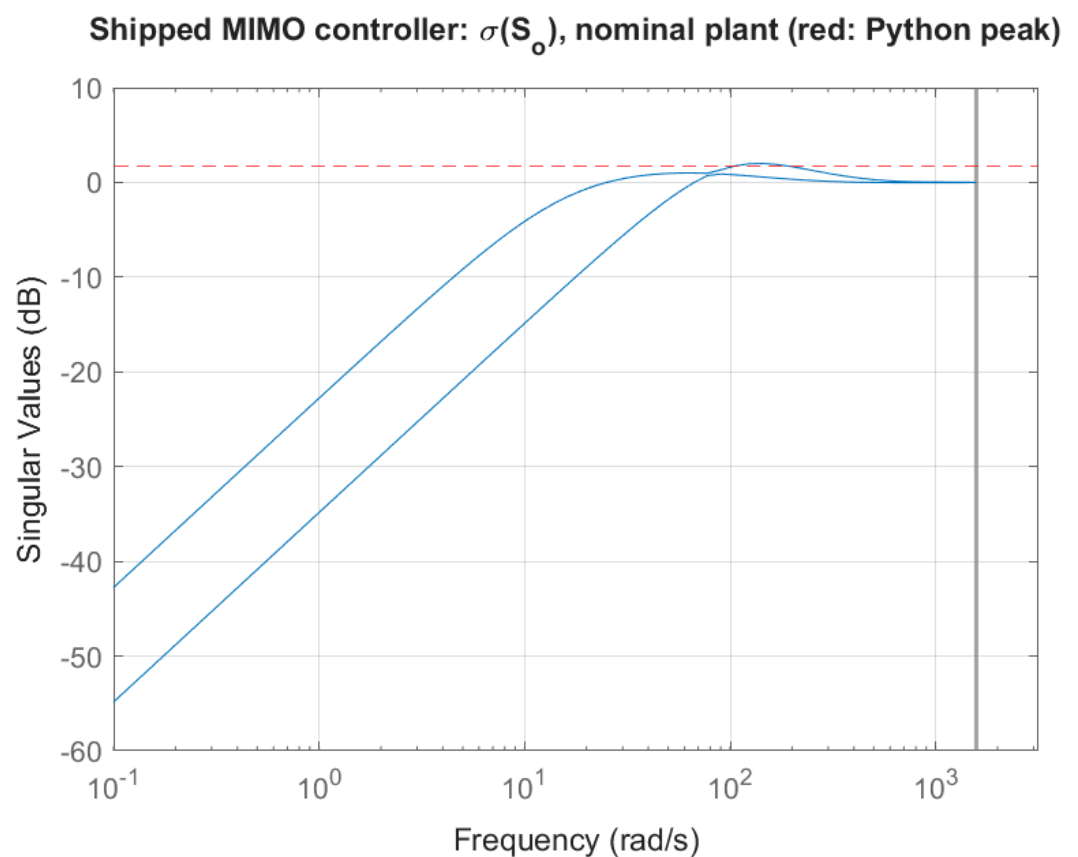


Figure 7. MATLAB cross-check of $\bar{\sigma}(S_o)$ versus frequency, built independently from the documented plant equations and the parsed coefficient header. Compare Figure 2.

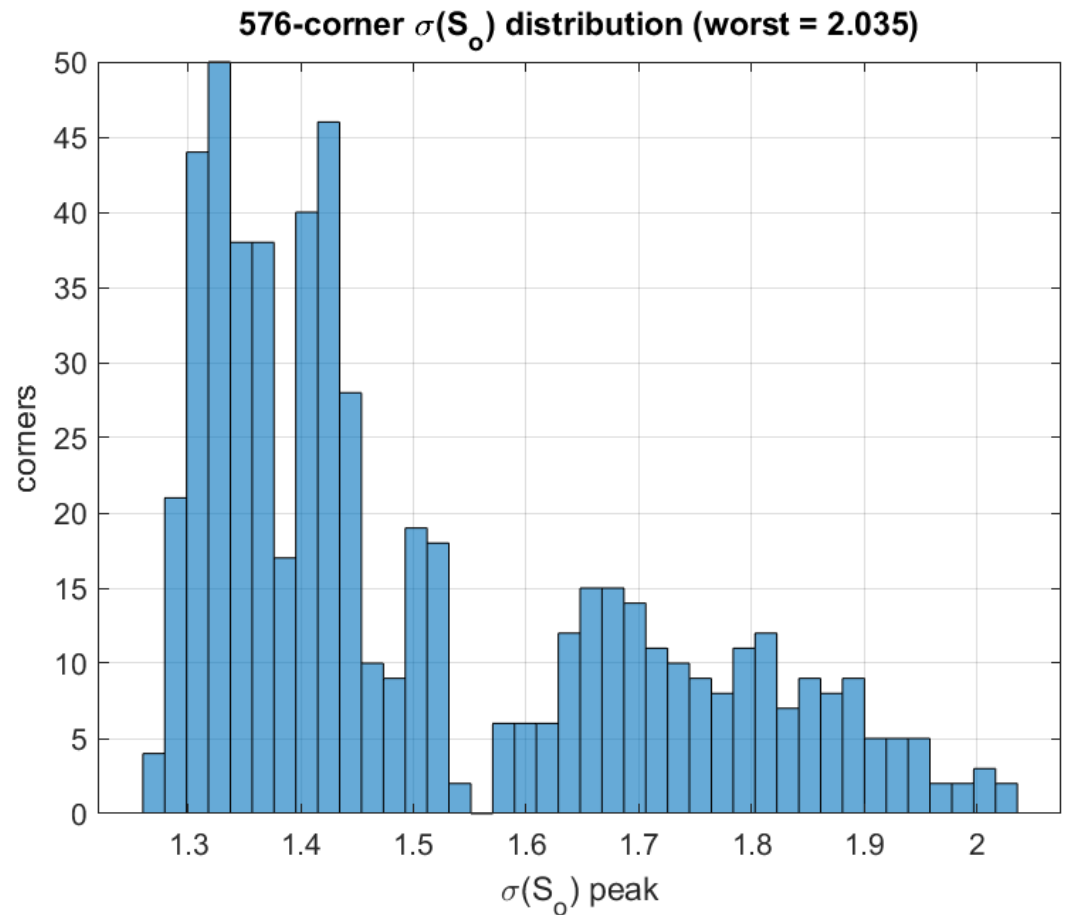


Figure 8. MATLAB cross-check of the per-corner robustness scatter (compare Figure 3). This sweep is the *full* 24×24 family at the nominal operating point, which is what surfaced the subsample correction of Section 6.

The cross-check earned its keep: it surfaced both corrections reported in Section 6, and both were then reproduced exactly by the Python model, so they are genuine properties of the design rather than tool disagreements. That is the ideal outcome for an independent implementation — it agrees everywhere the two tools should agree, and disagrees exactly where the original run was making an unstated approximation.

One residual limitation: the truth-model $\|\cdot\|_\infty$ check falls back to a dense singular value sweep, because the truth model mixes 0.12 rad s^{-1} vehicle dynamics with $\sim 10^5 \text{ rad s}^{-1}$ converter poles and the Hamiltonian bisection fails on that spread. A sweep is a lower bound — the conservative direction for a “does the truth model look worse” check — but not a proof. Design-plant against truth-model $\bar{\sigma}(S_o)$ at nominal agrees to 4.1 % (1.1606 vs. 1.2096) against a 20 % gate.

8. Conclusions and Future Work

- The coupling is real, one-directional, and operating-point dominated.** The droop ratio does not move the wheel speed ($G_{21} \equiv 0$, a modelling result with its residual measured at 0.44 % of the bus), so the plant is exactly upper triangular and the RGA is identically I at every frequency and corner. The live path is drive to share, through $\partial\alpha/\partial I_{tot} = -\Delta V_0 r_0 (1 - r_0) / (k_d I_{tot0}^2)$, at 39 % of the direct path at 2 A and $6.3\times$ it at 0.5 A. **The RGA is the wrong instrument for this plant**, which is itself a reportable finding.
- Decentralized control is justified, not merely convenient.** It is stable at all 4992 feasible Tier-1 corners, continuous and in the exact multirate discrete loop; it pays

- a bounded price in worst-corner sensitivity (2.55 vs. 1.92 on the same subsample); and the centralized controller's own gains say the same thing — it is about 98 % decentralized, with a 1.9 % coupling feedforward that the synthesis deliberately keeps small because it cannot trust the coupling's sign.
3. **The three headline metrics give three different answers**, and this is the result rather than an inconvenience: centralized wins worst-corner sensitivity, ties on regen bus excursion (both controllers are on the same current rail — that metric discriminates nothing), and wins or loses the drive-transient share excursion depending on the sign of ΔV_0 . Decentralized wins every saturated large-signal case.
 4. **The MIMO Youla-H $T(0) = I$ correction works and is cheap**. A constant right multiplier $M = [G_s(0)Y_H(0)]^{-1}$ on the Youla parameter generalizes the scalar gain-tuning recipe without touching poles or zero dynamics, reduces to it exactly in the 1×1 case, and combined with an exact rank-2 integrator split makes $T(0) = I$ structural rather than numerical — it survives truncation, discretization and float32 quantization ($\|T(0) - I\| = 7.3 \times 10^{-10}$; float32 replay error 1.79×10^{-6}).
 5. **Compute cost is not the deciding factor**, and in fact runs the counter-intuitive way: 28.5 against 49.5 kMAC/s, because the decentralized drive half cannot be a biquad cascade and needs a Hanus self-conditioned state-space form at a second rate. The genuine costs of centralization are halving the share loop rate, coupling the two control tasks' scheduling, and a matrix-valued anti-windup scheme.

Future work, in priority order.

(1) Measure ΔV_0 — a direct reading of two no-load bus setpoints that collapses the sign uncertainty bounding the entire centralized advantage. Nothing else here is worth more per unit of bench time; a re-synthesis at the measured sign could then justify a materially larger feedforward. **(2) Close the drive-channel calibration**: tire radius and encoder chain first (cheap, blocking for everything downstream), then k_t (requires selecting a motor), then τ_v and $T_{d,v}$ from a VESC current-step response, then the free-run current at 16 V which sets 93 % of b_{eff} . If calibration narrows the K_v envelope, re-run the drive weights with a lighter effort penalty and expect a materially faster drive channel. **(3) Address the K -envelope gap**: the shipped SISO controller was never gated at the light-load and FC-cruise points where K leaves $[0.55, 1.45]$; gain scheduling on I_{tot} would help both architectures, the centralized one more since it is the worse of the two there. **(4) Firmware integration is a separate, reviewed round**. The emitted coefficient header is deliberately not wired into the build, and on the evidence here there is no operational case for wiring it in before ΔV_0 is known.

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Data Availability Statement: All numbers in this report are generated artefacts of the internal controller_design_MIMO/ toolchain and are reproducible from the scripts and metric files referenced in [9–11].

Conflicts of Interest: The author declares no conflicts of interest.

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